

Additional file 1

The encoder, not the bins: decomposing peak-level MS/MS gains

Xulin Pan Chuanming Wang

This file reports the per-arm and per-seed results behind every figure in the main text, the full bootstrap distributions, and the peak-absorption analysis broken down by acquisition condition. Every table here is also provided as a machine-readable CSV in the repository under `results/`, and every number is regenerated from saved run artefacts by `scripts/make_supplementary.py` and `scripts/make_si_figures.py`.

S1 Per-arm, per-seed metrics

Table S1 gives every trained arm on both corpora. Twenty-seven training runs: five arms $\times$ three seeds on the open corpus, four $\times$ three on the single-library corpus. Parameter counts are identical between M1R and M1 by construction, and the binned baseline M0 carries 65% more parameters than either.

Every arm in both ladders is conditioned on the acquisition covariates (collision energy, adduct, polarity, instrument). In earlier versions of this work M0 alone was not: its encoder accepted the covariate vector and discarded it, so the binned baseline was the only arm predicting without knowing how the spectrum had been acquired. All six M0 rows above are refits under the corrected baseline and supersede the figures reported previously; the main text states what the correction cost the headline claim.

Conditioning was worth +0.0288 to M0 on the open corpus and +0.0142 on the single-library corpus. That the gain roughly doubles with the corpus’s acquisition diversity — 66 instrument types and 112 adducts against two and sixteen — is what one would expect if the covariates are doing the work we ascribe to them, and we note it as a consistency check rather than a result.

S2 Per-seed effect sizes and ordering checks

Each effect is the difference between two arms trained under the *same* seed, so the three values per row are matched rather than independent samples (Table S2). Where the main text claims an ordering between two effects, it is checked within each seed; Table S3 reports those checks.

Three things are worth reading off this table. The encoder-against-mass-axis row reverses on both corpora independently, which is why the main text declines to order those two anywhere. The encoder-against-aggregation row is consistent on both and negative on both, which is why aggregation is the one ranking the paper does claim. And the final row is why the main text ranks the mass axis against aggregation on the open corpus and declines to on the single-library corpus.

S3 Bootstrap distributions

Figure S1 shows the full distribution behind every interval reported in the main text: 5,000 resamples of the mean paired per-target difference, over targets, after averaging each target’s average

Table S1: Held-out metrics for every arm and seed. Macro AUPRC is the primary metric; chance is 0.0085 on the open corpus and 0.0093 on the single-library corpus. Machine-readable: `results/si_table_s1_arm_metrics.csv`.

corpus	arm	seed	parameters	best epoch	macro AUPRC	macro AUROC	Brier
open	M0	2026	1,172,752	13	0.1433	0.8815	0.00531
open	M0	7	1,172,752	14	0.1416	0.8793	0.00534
open	M0	13	1,172,752	14	0.1406	0.8788	0.00535
open	M1D	2026	714,528	12	0.1616	0.8930	0.00522
open	M1D	7	714,528	13	0.1569	0.8915	0.00527
open	M1D	13	714,528	14	0.1585	0.8940	0.00524
open	M1R	2026	712,336	14	0.1591	0.8991	0.00525
open	M1R	7	712,336	13	0.1570	0.8980	0.00525
open	M1R	13	712,336	14	0.1541	0.8960	0.00526
open	M1	2026	712,336	14	0.1731	0.9063	0.00513
open	M1	7	712,336	13	0.1709	0.9063	0.00517
open	M1	13	712,336	13	0.1687	0.9068	0.00514
open	M2	2026	829,584	14	0.1899	0.9072	0.00512
open	M2	7	829,584	14	0.1893	0.9052	0.00511
open	M2	13	829,584	12	0.1871	0.9090	0.00512
single library	M0	2026	1,172,752	14	0.0631	0.8096	0.00689
single library	M0	7	1,172,752	14	0.0621	0.8105	0.00688
single library	M0	13	1,172,752	13	0.0608	0.8109	0.00688
single library	M1R	2026	712,336	13	0.0867	0.8457	0.00673
single library	M1R	7	712,336	14	0.0856	0.8474	0.00672
single library	M1R	13	712,336	14	0.0840	0.8470	0.00675
single library	M1	2026	712,336	14	0.1119	0.8734	0.00654
single library	M1	7	712,336	13	0.1085	0.8717	0.00656
single library	M1	13	712,336	13	0.1162	0.8759	0.00652
single library	M2	2026	829,584	13	0.1460	0.8861	0.00640
single library	M2	7	829,584	13	0.1428	0.8824	0.00644
single library	M2	13	829,584	14	0.1444	0.8837	0.00641

Table S2: Effect sizes per seed, as differences in held-out macro AUPRC.

corpus	effect	seed 2026	seed 7	seed 13	mean	SD
open	interface (M1D – M0)	+0.0183	+0.0153	+0.0179	+0.0172	0.0016
open	attention (M1R – M1D)	−0.0025	+0.0001	−0.0045	−0.0023	0.0023
open	encoder (M1R – M0)	+0.0159	+0.0154	+0.0134	+0.0149	0.0013
open	mass axis (M1 – M1R)	+0.0140	+0.0139	+0.0147	+0.0142	0.0004
open	aggregation (M2 – M1)	+0.0168	+0.0184	+0.0184	+0.0179	0.0009
single library	encoder	+0.0236	+0.0235	+0.0232	+0.0234	0.0002
single library	mass axis	+0.0252	+0.0229	+0.0322	+0.0268	0.0049
single library	aggregation	+0.0341	+0.0343	+0.0282	+0.0322	0.0035

Table S3: Ordering checks. Each entry is the within-seed difference between two effects; an ordering is reported in the main text only where it holds in all three seeds.

corpus	comparison	seed 2026	seed 7	seed 13	mean	verdict
open	encoder – mass axis	+0.0019	+0.0015	−0.0012	+0.0007	reverses
open	encoder – aggregation	−0.0009	−0.0030	−0.0049	−0.0029	3/3 consistent
open	mass axis – aggregation	−0.0028	−0.0045	−0.0037	−0.0037	3/3 consistent
single library	encoder – mass axis	−0.0016	+0.0006	−0.0090	−0.0033	reverses
single library	encoder – aggregation	−0.0105	−0.0108	−0.0049	−0.0087	3/3 consistent
single library	mass axis – aggregation	−0.0089	−0.0114	+0.0041	−0.0054	reverses

precision across the three seeds. Table S4 summarises them.

Table S4: Bootstrap summaries. Machine-readable draws: `results/si_table_s3_bootstrap_draws.csv.gz`.

corpus	effect	mean	SD	2.5%	97.5%
open	interface	0.01727	0.00360	0.01020	0.02419
open	attention	−0.00230	0.00221	−0.00665	0.00196
open	encoder	0.01498	0.00362	0.00798	0.02210
open	mass axis	0.01415	0.00230	0.00969	0.01865
open	aggregation	0.01785	0.00222	0.01360	0.02233
single library	encoder	0.02344	0.00194	0.01972	0.02729
single library	mass axis	0.02678	0.00243	0.02208	0.03168
single library	aggregation	0.03219	0.00243	0.02751	0.03696

What these intervals do and do not cover

Each interval resamples targets, having first averaged average precision over the three seeds. It therefore describes uncertainty from the finite set of targets, not from initialisation. Seed variability is reported separately as the SD column of Table S2, and the two should be read together: on the single-library corpus the mass-axis effect has a seed SD of 0.0049, comparable to the width of its bootstrap interval, which is the statistical form of the ordering failure in Table S3. A variance-components treatment with targets and seeds as crossed random effects would fold both into one interval and we regard it as the right analysis for a larger replicate count.

The attention arm

M1D removes self-attention from every block while holding the mass axis, the peak tokens and the acquisition conditioning fixed, so that M1D – M0 prices the interface and M1R – M1D prices attention. Removing attention frees 66,048 weights per block; the feed-forward is widened to 548 units in this arm alone so that the control is not also the smaller model (714,528 parameters against 712,336). The attention effect is the only one in this work whose interval includes zero, and it is also the only one positive in fewer than three seeds of three.

S4 Per-target average precision

`results/si_table_s2_per_target_ap.csv` gives average precision for every scored target, for every arm, both as the seed-averaged value used in the paired analysis and per seed: 433 targets on the

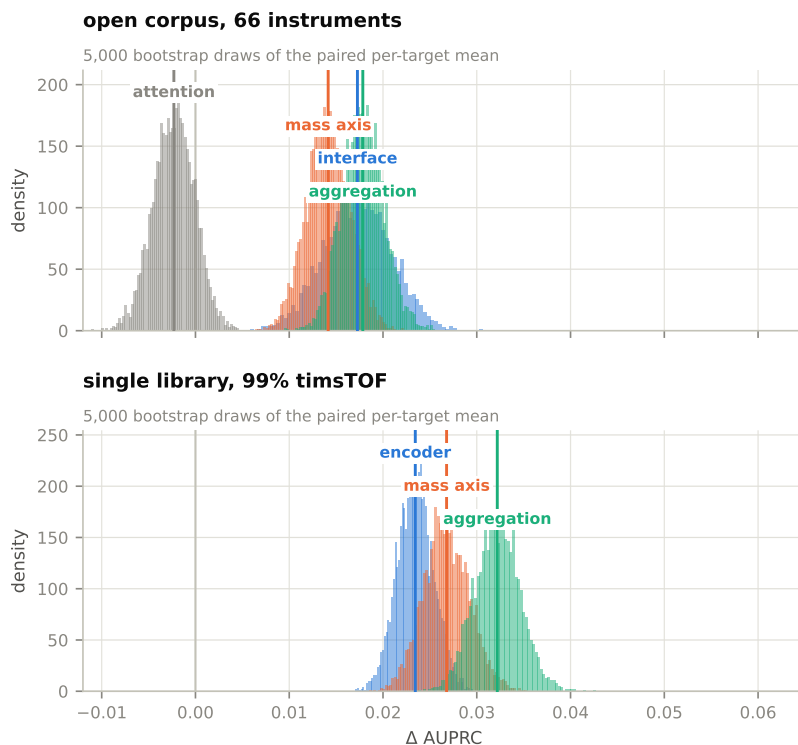


Figure S1: Bootstrap distributions on each corpus. The open panel shows the encoder split into its two components: the interface distribution sits far from the others while the attention distribution is centred on zero. On the single-library corpus, which has no attention arm, the composite encoder effect is shown instead. The encoder and mass-axis distributions overlap heavily there, and the mass-axis and aggregation ones likewise, which is the same fact Table S3 records as two sign reversals; only the encoder and aggregation distributions separate.

open corpus and 477 on the single-library corpus, 910 rows in total. All paired differences in the main text are recoverable from this file by subtraction.

S5 Absorption by acquisition condition

The main text reports that 23.6% of the open corpus’s held-out peaks fall into a 0.5 Da bin already occupied. Figure S2 breaks that down by adduct, collision energy and instrument family.

Absorption at 0.5 Da by acquisition stratum, open corpus held-out spectra

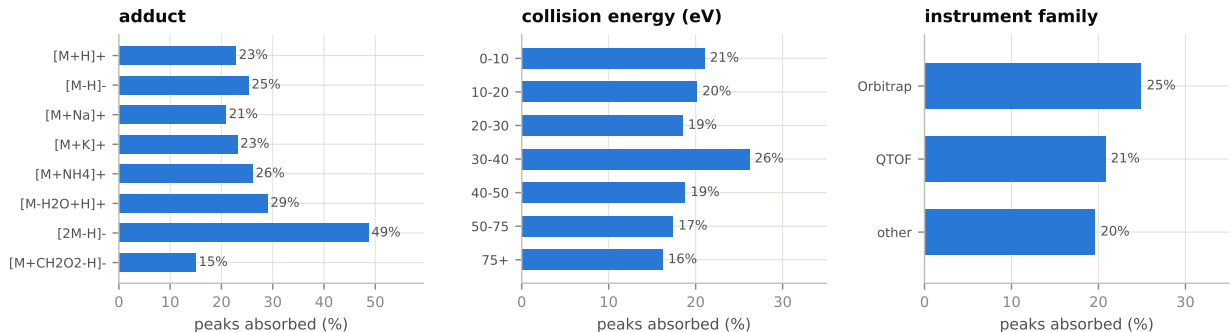


Figure S2: Absorption at 0.5 Da by acquisition stratum, over the open corpus’s held-out spectra. Strata with fewer than 20,000 peaks are omitted.

Absorption tracks peak density rather than any chemical property of the adduct. The highest rate among common adducts is $[2M-H]^-$ at 48.7%, whose spectra are dimeric and correspondingly crowded; $[M+H]^+$, which supplies 62% of held-out spectra, sits at 22.8%. Across collision energies the range is 16% to 26% with no monotone trend. Orbitrap spectra lose 24.9% against QTOF’s 20.8%, again following density (52.6 against 47.2 peaks per spectrum) rather than resolving power.

This is worth stating because it constrains the interpretation in the main text. Binning costs more where peaks are dense, and density varies with acquisition conditions that have nothing to do with how much mass precision the instrument actually delivered.

Machine-readable, all under `results/`:

```
si_absorption_by_adduct.csv
si_absorption_by_collision_energy.csv
si_absorption_by_instrument_family.csv
```

S6 Bin-width sweep

Table S5 gives the sweep in full. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. One run per width, on the single-library corpus.

The absorption column here is computed over the competition test spectra, which are denser than the corpus’s own held-out spectra (72.5 against 58 peaks per spectrum); the corresponding held-out figures are 34.4% at 0.5 Da falling to 6.5% at 0.01 Da, and those are the numbers the main text uses.

Table S5: Bin-width sweep, single seed, single-library corpus.

bin width (Da)	peaks absorbed	macro AUPRC	share of the 0.5 Da gap closed
0.5	37.5%	0.0867	0%
0.25	36.8%	0.0818	-19%
0.1	33.1%	0.0801	-26%
0.05	25.9%	0.0909	17%
0.01	7.5%	0.1030	65%
full precision	0.0%	0.1119	100%

S7 Reproduction

Software: Python 3.10, PyTorch 2.11.0+cu128, RDKit 2026.03.6, NumPy 2.4.4, pandas 2.3.3, pyarrow 24.0.0. Exact pins are in `requirements.txt`.

The full path from the corpus file to every number in this file is

```
python scripts/make_open_subset.py --root .
python -m dbf2 prepare --root . --dataset open
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15 --seed 7
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15 --seed 13
python scripts/make_supplementary.py
python scripts/make_si_figures.py
```

with the same three commands minus `--dataset open` for the single-library corpus, and

```
for w in 0.25 0.1 0.05 0.01; do
    python -m dbf2 train --root . --rung M1 \
        --quantise $w --tag W$w --epochs 15
done
```

for the sweep. `python -m pytest src/dbf2/tests` runs 50 unit tests, 14 of which cover the rounding control, including that M1R and M1 have identical parameter counts and that their configurations differ in exactly one field.

A note on what the subset builder reads

`make_open_subset.py` selects the GNPS, MassBank and MoNA rows out of the aggregated CASMI 2026 file. Those spectra are openly licensed at their sources, but reproducing this work as written still requires that aggregated file. Fetching the records from GNPS, MassBank and MoNA directly is recorded as an open item in the repository. `docs/corpus_composition.md` reports each source library’s contribution and states which parts are redistributable.